BRIGHTLEARN TUTORIALS

DATA SCIENCE

EXERCISE 1 : SQL FUNDAMENTALS

Database : employee_db

1. SELECT statement

SQL query to retrieve all columns from the employees table

Syntax :

```
SELECT *
FROM employees;
```

Output :

id	first_name	last_name	department	salary	hire_date	city
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San Francisco
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San Francisco

2. SELECT DISTINCT Statement

SQL query to find all unique departments

Syntax :

```
SELECT DISTINCT department
FROM employees;
```

Output :

department
IT
HR
Finance
Marketing

3. ORDER BY statement

SQL query to retrieve all employees first and last name, ordered by salary in descending order

Syntax :

```
SELECT first_name,
       last_name, Salary
FROM employees
ORDER BY salary DESC;
```

Output :

First_name	last-name	Salary
Emily	Davis	62000
Mike	Johnson	60000
Robert	Wilson	59000
John	Doe	55000
Sarah	Brown	53000
Daniel	Clark	53000
David	White	52000
Jessica	Moore	51000
Laura	Hall	50000
Jane	Smith	48000

4. LIMIT Statement

SQL query to retrieve top 5 highest paid employees

Syntax :

```
SELECT first_name,
       last_name,
       salary
FROM employees
ORDER BY salary DESC
LIMIT 5;
```

Output :

first_name	last_name	salary
Emily	Davis	62000
Mike	Johnson	60000
Robert	Wilson	59000
John	Doe	55000
Sarah	Brown	53000

5. WHERE Statement

SQL query to find employees who work IT department

Syntax :

```
SELECT first_name,
       last_name
FROM employees
WHERE department = 'IT';
```

Output :

first_name	last_name
John	Doe
Sarah	Brown
Emily	Davis
Laura	Hall

6. AND statement

SQL query to find employees who work in Finance department AND have a salary greater than 58000

Syntax :

```
SELECT first_name,
       last_name,
       salary
FROM employees
WHERE department = 'Finance' AND
      salary > 58000;
```

Output :

first_name	last_name	salary
Mike	Johnson	60000
Robert	Wilson	59000

7. OR statement

SQL query to find employees who work in the HR OR Marketing department

Syntax :

```
SELECT first_name, last_name,
       department
FROM employees
WHERE department = 'HR' OR department =
  'Marketing';
```

Output :

first_name	last_name	department
Jane	Smith	HR
Jessica	Moore	HR
David	White	Marketing
Daniel	Clark	Marketing

8. NOT statement

SQL query to find employees who do not work in IT department

Syntax :

```
SELECT first_name,
       last_name,
       department
FROM employees
WHERE department != 'IT';
```

Output :

first_name	last_name	department
Jane	Smith	HR
Mike	Johnson	Finance
David	White	Marketing
Robert	Wilson	Finance
Jessica	Moore	HR
Daniel	Clark	Marketing

9. IN statement

SQL query to find employees who are in HR, IT or Finance department

Syntax : SELECT first_name,
last_name,
department
FROM employees
WHERE department IN ('HR', 'IT', 'Finance');

Output :

first_name	last_name	department
John	Doe	IT
Jane	Smith	HR
Mike	Johnson	Finance
Sarah	Brown	IT
Emily	Davis	IT
Robert	Wilson	Finance
Jessica	Moore	HR
Laura	Hall	IT

10. Combining Conditions

SQL query to find employees in IT department, have salary > 50000 and located in New York

Syntax :

```
SELECT first_name, last_name,
       department, salary, city
FROM employees
WHERE department = 'IT' AND salary > 50000
      AND city = 'New York';
```

Output :

first_name	last_name	Salary	city
John	Doe	55000	New York
Sarah	Brown	53000	New York

11. Combining WHERE, AND, ORDER BY

SQL query to retrieve first and last names of employees working in Finance or Marketing department, earn more than 52000 and order results by salary in descending order

Syntax:

```
SELECT first-name, last_name,
        department, salary
FROM employees
WHERE department IN ('Finance', 'Marketing')
 AND salary > 52000
ORDER BY salary DESC;
```

Output:

first-name	last_name	department	salary
Mike	Johnson	Finance	60000
Robert	Wilson	Finance	59000
Daniel	Clark	Marketing	53000

12. Combining SELECT DISTINCT, WHERE AND IN

SQL query to find all unique cities employees work, excluding those in IT and HR department

Syntax:

```
SELECT DISTINCT city
FROM employees
WHERE department NOT IN ('IT', 'HR');
```

Output:

City
Los Angeles
San Francisco
Houston
Chicago

13: Combining WHERE, NOT, AND and ORDER BY
SQL query to retrieve employees who are NOT in Finance department, salary > 50000 and order results by hire date ascending order

Syntax:

```
SELECT first_name, last_name,
       salary, hire_date
FROM employees
WHERE department != 'Finance'
  AND salary > 50000
ORDER BY hire_date ASC;
```

Output:

first_name	last_name	Salary	hire_date
Emily	Davis	62000	2015-02-14
David	White	52000	2016-04-10
John	Doe	55000	2018-06-15
Jessica	Moore	51000	2018-05-22
Sarah	Brown	53000	2021-03-25
Daniel	Clark	53000	2022-06-01

14. Combining WHERE, OR, IN and LIMIT
SQL query to find first 3 employees working in either Chicago or Los Angeles and belong to IT or Marketing department

Syntax:

```
SELECT first_name, last_name,
       department, city
FROM employees
```

```
WHERE city IN ('Chicago', 'Los Angeles')
  AND department IN ('IT', 'Marketing')
LIMIT 3;
```

Output :

first_name	last_name	department	city
Emily	Davis	IT	Chicago
Daniel	Clark	Marketing	Chicago